

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Quiz - 3
 Duration: 20 minutes

Semester: Summer 2026
 Full Marks: 20

CSE 340: Computer Architecture

Name: <u>Solution</u>	ID:	Section:
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1. Suppose we run the following instruction set in a **pipelined datapath**. Find the number of Data hazards in each line. [1 x 11]

Instruction Set	Number of Data Hazard(s)
add x7, x1, x2	0
ld x5, 0(x7)	1
sub x9, x7, x3	1
and x6, x5, x9	2
addi x11, x6, 2	1
ori x10, x11, 7	1
sd x10, 8(x10)	2
xor x12, x11, x6	0
sll x13, x12, x10	1
sd x13, 16(x11)	1
or x15, x13, x12	1

2. The following instruction will be run in a **single cycle datapath**. IF = 10s, ID = 10s, EX = 20s, MEM = 10s, WB = 10s.

Answer the following questions. [2+3+2+2]

Instruction set:

add x5, x6, x7 ——— 50ns
 sub x5, x7, x8 ——— 50ns
 ld x8, 12(x5) ——— 60ns
 slli x8, x8, 3 ——— 50ns
 srli x18, x8, 4 ——— 50ns
 sd x8, 10(x5) ——— 50ns

- a. How many clock cycles are needed to run the following instruction? Answer: 6
 b. What would be the clock period of the system? Answer: 60ns
 c. How many data hazards are there? Answer: 0
 d. Find the total time to run the instruction set. Answer: (6x60) = 360ns